

NeuroGuardian X

Project overview, input data, output data, architecture, current status, and release-readiness report.

Prepared date	September 2, 2026
App version	0.3.0+3
Current stage	Hardware-connected prototype moving toward real-world pilot testing
Primary safety goal	Detect falls and no-movement emergencies, then alert guardians with GPS and nearby hospital context

EXECUTIVE POSITION

NeuroGuardian X is an emergency-support and risk-detection system, not a medical diagnosis device. The safest path to society-scale use is a phased pilot: real sensors, calibrated thresholds, verified guardian notifications, emergency-service partnerships, and formal privacy/medical review.

1. Project Overview

NeuroGuardian X combines an ESP32-S3 wrist-worn device, a Flutter Android app, and a small emergency backend. The watch streams movement and health signals to the phone over Bluetooth Low Energy. The app detects risk patterns, starts a 30-second fall countdown, lets the user cancel with an I AM OKAY control, and sends emergency alerts when the situation remains critical.

- Target users: older adults, workers, patients under observation, and people who may need rapid guardian awareness after a fall.
- Main value: faster response during fall, no-movement, sudden heart-rate drop, low oxygen, stress, temperature, or manual SOS conditions.
- Important boundary: the system supports emergency awareness; it should not claim heart attack confirmation, diagnosis, or replacement for a doctor.

2. Current Build Status

Area	Current implementation	Real-world role
Flutter app	Android APK installed for phone testing; includes dashboard, device, SOS, care, history, and settings flows.	Main user interface and emergency control layer.
BLE connection	App scans for NeuroGuardianX and listens for JSON packets from the ESP32-S3 wearable.	Live connection between wrist hardware and phone.
Fall protocol	Fall signal starts a 30-second timer; movement cancels it; critical fall can trigger immediate SOS.	Core safety automation.
Backend	FastAPI backend keeps Google/WhatsApp keys out of app code and prepares alert payloads.	Secure notification and hospital lookup layer.

Area	Current implementation	Real-world role
Firmware	ESP32-S3 firmware has real-sensor mode, IMU support, optional MAX30102/GPS flags, buzzer, vibration, button, and battery hooks.	Wearable data producer.

3. System Architecture

Layer	Input	Processing	Output
ESP32-S3 watch	IMU, heart sensor, SpO2, GSR, temperature, ECG, GPS, button, battery	Reads sensors, detects impact/no movement, streams compact JSON over BLE	Vitals packets, fall events, buzzer/vibration feedback
Flutter app	BLE packets, user settings, guardian number, phone GPS	Risk detection, fall countdown, movement cancellation, care suggestions, alert history	Live dashboard, I AM OKAY reset, SOS request
Emergency backend	SOS payload, location, guardian target, API keys stored in .env	Hospital lookup, message formatting, WhatsApp/push routing	Guardian alert with Google Maps link and top medical facilities

ARCHITECTURE FLOW

ESP32-S3 Watch -> BLE JSON -> Flutter App -> 30-sec Safety Logic -> Backend -> Google Places -> WhatsApp / Push / SMS fallback

4. Input Data

4.1 Wearable Sensor Inputs

Input	Suggested source	Purpose	Current status
Acceleration and gyro	MPU6050 now; BMI270/ICM-42688 recommended for production	Fall detection, movement recovery, activity context	Firmware hooks present
Heart rate	MAX30102/MAX86141 optical PPG	Sudden HR drop, cardiac load, stress support	Optional firmware flag
SpO2	MAX30102/MAX86141 optical PPG	Low oxygen risk detection	Optional sensor calibration needed
GSR/EDA	Skin conductance sensor	Stress response detection	Analog input present
Body temperature	MAX30205/MAX30208 or skin temperature sensor	Fever/heat strain context	App meter present
ECG amplitude	AD8232 or compact ECG front-end	Future ECG model input	Analog input present; validation needed
GPS	Watch GPS or phone GPS fallback	SOS location and nearby facility lookup	Payload support present
Battery/manual SOS	Voltage divider/fuel gauge and watch button	Battery warning and user-triggered alert	Firmware inputs present

4.2 User, Backend, and Dataset Inputs

- Guardian WhatsApp number is entered by each wearable owner; it is not hard-coded.

- Backend URL, device token, and push target connect the app to the alert service.
- Google Maps and WhatsApp provider secrets stay in backend .env, not in Flutter app code.
- Doctor-suggested medicines are stored only when entered; the app should not automatically prescribe medicine.

Dataset	Signal focus	Planned use
MIT-BIH Arrhythmia Database	ECG rhythm examples	Arrhythmia research model training and validation.
Sudden Cardiac Death Holter Database	Long ECG records around serious cardiac events	High-risk ECG feature study; not a sole emergency rule.
Wearable Device Dataset	BVP, EDA, temperature, motion	Stress/activity features similar to watch input.
Wearable Exam Stress Dataset	Stress-related wearable signals	Stress trend validation and false-alert reduction.
Prototype watch logs	Actual device packets	Most important source for calibration and field performance.

5. Output Data

Output	Destination	Meaning
Live vitals	Dashboard	HR, SpO2, HRV, stress, GSR, ECG, body temperature, activity, movement, battery, GPS state.
AI risk detection label	Dashboard	User-friendly condition label such as minor stress, high temperature, low oxygen, or critical fall.
Fall countdown	Dashboard and SOS tab	30-second timer after fall; movement cancels it; no movement triggers SOS.
Care suggestions	Care tab	Exercise/home-remedy style guidance for minor criteria; serious criteria shows consult doctor.
Guardian alert	WhatsApp/push/SMS fallback	Reason, latest vitals, exact Google Maps link, and top nearby medical facilities.
Wearable feedback	Watch buzzer/vibration/LED	Immediate physical feedback during emergency countdown and connection events.

6. Emergency Logic

- ESP32-S3 detects fall impact or sends a fall_detected event.
- The app starts a 30-second countdown with vibration pulses every 2 seconds and BUZZER_ON.
- If movement returns, the app stops the timer and sends BUZZER_OFF.
- If no movement remains for 30 seconds, the app fetches GPS and sends SOS.
- If fall plus sudden heart-rate drop is detected, immediate SOS triggers without waiting.

7. Hardware and Firmware Architecture

Subsystem	Recommended part	Reason
Controller	ESP32-S3 module	BLE support, enough processing power, common dev-board ecosystem.
Motion	BMI270/ICM-42688 for production; MPU6050 for prototype	Better fall and movement detection with calibrated acceleration/gyro data.
Optical vitals	MAX86141/MAX30102	Heart rate and SpO2 support for wrist prototypes.
Temperature	MAX30205/MAX30208 or skin temperature sensor	Body temperature trend input.
Stress	GSR/EDA electrodes	Stress trend signal when combined with HRV and movement.
ECG optional	AD8232 or medical-grade ECG front-end	Research input only unless clinically validated.
Location	Phone GPS fallback plus optional watch GPS	Best reliability because indoor GPS can fail on small wearables.
Feedback/power	Vibration, buzzer, LED, LiPo, charger IC, fuel gauge	Local warning plus reliable battery health monitoring.

8. Release Readiness

BIGGEST RISK

The product must not miss a real emergency and must not create frequent false emergencies. This is the central safety problem to solve before public release.

Risk	Why it matters	Control needed
False fall alert	Can worry guardians or waste emergency resources.	Test with walking, running, sitting, stairs, phone drops, and sleep movement.
Missed fall	Can delay help when the user is unconscious.	Tune IMU thresholds using real falls/safe simulations and multiple body positions.
Loose sensor contact	Wrist vitals can be inaccurate.	Add signal-quality checks and show warnings when readings are unreliable.
Indoor GPS failure	Location may be poor inside buildings.	Use phone GPS, last-known location, Wi-Fi/network location, and accuracy labels.
Notification failure	Guardian may not receive message.	Use WhatsApp, push fallback, SMS composer fallback, retry logging, and delivery status.
Medical/legal exposure	Public health products require careful claims.	Use disclaimers, consent, privacy policy, field trials, and clinical/regulatory advice.

9. Next Milestones

- Wait for Meta WhatsApp template approval, then test automatic WhatsApp alerts.
- Wire final sensors and enable firmware flags after libraries and wiring are verified.
- Run controlled fall/no-movement tests and tune thresholds against false alarms.
- Collect prototype logs and train AI models using PhysioNet plus real watch data.

- Prepare privacy policy, user consent, emergency-contact onboarding, and pilot-study documentation.

10. Local Development References

Item	Location
Flutter app	C:\Users\hpl\Downloads\neuro guardian x\neuroguardian_app
Android APK	C:\Users\hpl\Downloads\neuro guardian x\neuroguardian_app\build\app\outputs\flutter-apk\app-debug.apk
ESP32 firmware	C:\Users\hpl\Downloads\neuro guardian x\neuroguardian_app\firmware\esp32_neuroguardian_ble\esp32_neuroguardian_ble.ino
Backend server	C:\Users\hpl\Downloads\neuro guardian x\neuroguardian_app\backend\emergency_server.py
Backend secrets	C:\Users\hpl\Downloads\neuro guardian x\neuroguardian_app\backend\.env (do not publish)
AI model plan	C:\Users\hpl\Downloads\neuro guardian x\neuroguardian_app\docs\physionet_model_training_plan.md